

UNIVERSIDADE D
COIMBRA

Tiago Silva

Towards Fair Algorithmic Recourse with Kolmogorov-Arnold Networks

Dissertation in the context of the Master in Informatics Engineering, specialization in Artificial Intelligence, advised by Professor Pedro Henriques Abreu and Professor Bruno Veloso and presented to the Department of Informatics Engineering of the Faculty of Sciences and Technology of the University of Coimbra.

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Rumo a um Recurso Algorítmico Justo com Redes de Kolmogorov-Arnold

Dissertação no âmbito do Mestrado em Engenharia Informática, especialização em Inteligência Artificial, orientada pelo Professor Pedro Henriques Abreu e pelo Professor Bruno Veloso e apresentada ao Departamento de Engenharia Informática da Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

Janeiro 2027

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I would like to thank...

Statement on the use of artificial intelligence tools

This document was reviewed and refined with the assistance of Large Language Models, such as ChatGPT, or similar tools, which helped check grammar, correct typos, and enhance clarity. Any Artificial Intelligence-generated text or content is clearly indicated within the document and is used only to a limited extent. The overall content and ideas remain solely the responsibility of the author.

Abstract

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Chapter 1

Introduction

1.1 Context and Motivation

1.2 Objectives and Research Questions

1.3 Work Plan and Risk Analysis

1.3.1 Work Plan

1.3.2 Risk Analysis and Mitigation

1.4 Document Structure

Chapter 2

State of the Art

2.1 Background Knowledge

2.1.1 Notation

This subsection fixes the notation used in the rest of the document. Uppercase italic letters denote random variables, such as a feature X_j, the outcome Y or the protected attribute A, and the corresponding lowercase letters denote the values they take (x_j, y, a). Bold letters denote vectors or sets of variables: $\mathbf{X} = (X_1, \ldots, X_d)$ collects the d features and $\mathbf{x}$ holds the feature values of one person. Calligraphic letters denote graphs and sets, such as the causal graph $\mathcal{G}$, the set $\mathcal{I}$ of intervened variables and the set $\mathcal{F}(\mathbf{x}^{\mathrm{F}})$ of feasible actions of a person; $\mathcal{N}$ and Unif denote the normal and uniform distributions. The index j refers to the variable being explained, i, k and l to its parents or to intervened variables, m to a basis function and t to a time window. A hat marks a quantity estimated from data, as in $\hat{f}_j$ for the estimate of f_j. The letter σ has two uses, kept apart by their form: $\sigma(z)$, always written with an argument, is the logistic function, and σ_j^2, always written with a subscript, is the noise variance of X_j.

Table 2.1 lists the symbols that are used in more than one section, together with the section that defines each of them. Symbols used in a single place are defined where they appear.

Symbol	Meaning	Section
People, decisions and recourse		
$\mathbf{x} = (x_1, \ldots, x_d)$	feature values of one person	2.1.3
X_j, Y	j-th feature; outcome	2.1.3
h	classifier (decision model)	2.1.3
$\mathbf{x}^{\mathrm{F}}$	observed (factual) feature values of one person	2.1.3
$\mathbf{x}^{\mathrm{CFE}}$	counterfactual explanation of $\mathbf{x}^{\mathrm{F}}$ (found without a causal model)	2.1.3
$\boldsymbol{\theta}_{\mathcal{I}}$	action $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$ taken by a person	2.1.3
$c(\mathbf{x}^{\mathrm{F}}, \boldsymbol{\theta}_{\mathcal{I}})$	cost of the action $\boldsymbol{\theta}_{\mathcal{I}}$ for the person $\mathbf{x}^{\mathrm{F}}$	2.1.3
A, a	protected attribute and its value	2.1.4
Causal models		
$\mathcal{G}$	causal graph	2.1.2
$\mathbf{PA}_j$	parents of X_j	2.1.2
f_j	structural assignment (equation) of X_j	2.1.2
U_j, U_Y	noise variables of X_j and of Y	2.1.2
$\mathrm{do}(X_i = \theta)$	intervention that sets X_i to θ; $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$ sets every X_i with $i \in \mathcal{I}$ to θ_i	2.1.2
$\mathbf{u}^{\mathrm{F}}$	noise values inferred from the factual features $\mathbf{x}^{\mathrm{F}}$	2.1.3
$\mathbf{x}_{\boldsymbol{\theta}_{\mathcal{I}}}(\mathbf{u}^{\mathrm{F}})$	counterfactual: the person's features had $\mathbf{X}_{\mathcal{I}}$ been set to $\boldsymbol{\theta}$	2.1.3
$\mathbf{v}_a(\mathbf{u}^{\mathrm{F}})$	counterfactual twin: all of the person's variables, $\mathbf{v}^{\mathrm{F}} = (\mathbf{x}^{\mathrm{F}}, a^{\mathrm{F}})$, had A been a	2.1.4
$\sigma(z)$	logistic function, $1/(1 + e^{-z})$	2.1.2
Change over time		
t, P_t	time window; distribution of the data in window t	2.1.5
Curves and the causal KAN		
B_m, β_m	m-th B-spline basis function and its height (coefficient)	2.1.6
α_j	constant term in the equation of X_j	2.1.6
ϕ_{ij}	curve on the edge $X_i \to X_j$	2.1.6
ψ_{klj}	surface for the pair of parents X_k and X_l of X_j	2.1.6
λ	strength of the smoothness belief	**??**
σ_j^2	noise variance of X_j	**??**

Table 2.1: Symbols used in more than one section, with the section that defines each.

2.1.2 Structural Causal Models and Interventions

Recourse asks what would happen to a person's decision if they acted, and counterfactual fairness asks what the decision would have been had a protected attribute been different. Both are, inevitably, causal questions, and this thesis answers them with structural causal models (SCMs) [Pearl, 2009a; Peters et al., 2017]. This subsection introduces SCMs and interventions; counterfactuals computed from an SCM follow in Section 2.1.3, where they are first needed.

Structural causal models. An SCM describes how each variable is produced from its direct causes. Following Peters et al. [2017], an SCM over $X_1, \ldots, X_d$ consists of d structural assignments

$$X_j := f_j(\mathbf{PA}_j, U_j), \qquad j = 1, \ldots, d, \tag{2.1}$$

where $\mathbf{PA}_j \subseteq \{X_1, \ldots, X_d\} \setminus \{X_j\}$ are called the parents of X_j, together with a joint distribution $P_{\mathbf{U}}$ over the noise variables $U_1, \ldots, U_d$, which are required to be jointly independent. Peters et al. [2017] write the noise variables as N_j. This thesis writes U_j, the letter used by Pearl [2009b] and the recourse literature [Karimi et al., 2021], where U_j is called an exogenous variable and X_j an endogenous one. Drawing an arrow from each parent to its child gives the causal graph $\mathcal{G}$, which Peters et al. [2017] assume to be acyclic. In Pearl's formulation and in the recourse literature, independent noise is not part of the definition but an extra assumption, which this thesis makes: Pearl [2009b] calls a model Markovian when its graph is acyclic and its exogenous variables are jointly independent.

The parents of X_j are its direct causes, and U_j gathers everything else that affects X_j but is not modelled. Requiring the noise variables to be independent therefore rules out hidden confounders, that is, unobserved variables that affect two or more of the modelled ones; this assumption is called causal sufficiency. An acyclic SCM entails a single joint distribution over $X_1, \ldots, X_d$ [Peters et al., 2017], and in a Markovian model this distribution factorises into one conditional distribution per variable [Pearl, 2009a,b]:

$$P(X_1, \ldots, X_d) = \prod_{j=1}^{d} P(X_j \mid \mathbf{PA}_j). \tag{2.2}$$

Each factor is the causal mechanism of one variable. The principle of independent mechanisms states that these factors do not inform or influence one another, so that one of them can change while the others stay the same [Peters et al., 2017]. Without causal sufficiency, Equation 2.2 no longer holds in general. One solution is to add the hidden confounder to the model as a latent variable. Louizos et al. [2017], for example, infer an unobserved confounder Z from observed proxies of it, assuming that the proxies carry enough information to recover the joint distribution of Z and the observed variables; since Z is the only hidden confounder in their model, adding it makes the model Markovian again.

Running example. Figure 2.1 shows the SCM used as a running example. A person's income X_1 affects their savings X_2, and both affect the bank's decision Y, where $Y = 1$ means that the loan is approved:

$$\begin{aligned} X_1 &:= U_1, \\ X_2 &:= 0.5\, X_1^2 + U_2, \\ Y &:= \mathbf{1}[\, U_Y < \sigma(-4 + X_1 + 0.5\, X_2)\,], \end{aligned} \tag{2.3}$$

with $U_1 \sim \text{Unif}(0, 4)$, $U_2 \sim \mathcal{N}(0, 0.3^2)$ and $U_Y \sim \text{Unif}(0, 1)$, where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function and $\mathbf{1}[\cdot]$ equals 1 when its condition holds and 0 otherwise. The last equation means that a person is approved with probability $\sigma(-4 + X_1 + 0.5\, X_2)$. The noise U_2 can be read as a saving habit: a positive value means that the person saves more than others with the same income.

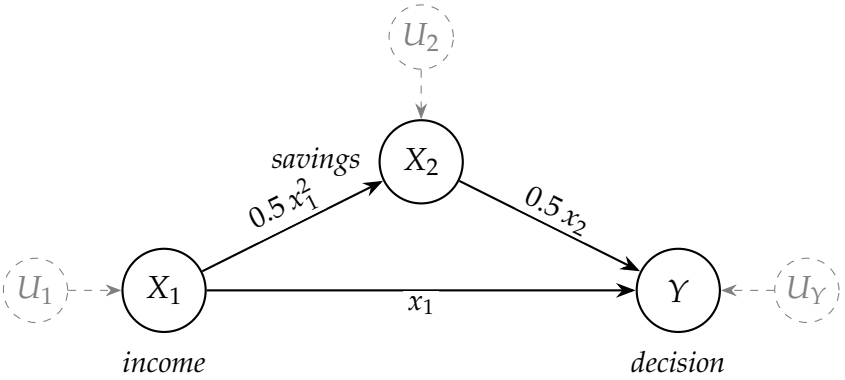
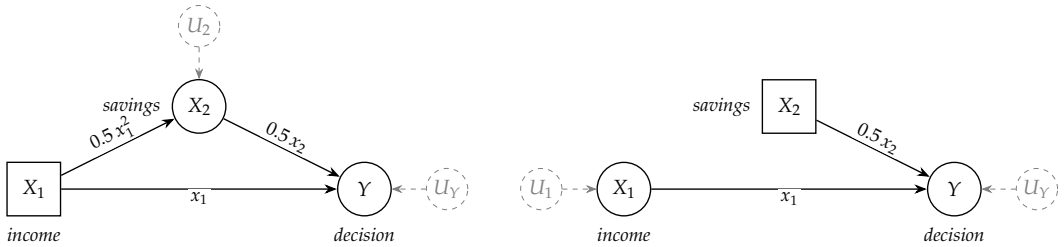


Figure 2.1: Causal graph of the running example (Equation 2.3). Solid arrows are causal effects, labelled with the term that each parent adds to its child's equation (for Y, to the argument of σ); dashed arrows come from the noise variables.

Interventions. An intervention sets a variable from outside the system. The intervention $\mathrm{do}(X_i = \theta)$ replaces the equation of X_i with the constant θ and leaves every other equation unchanged [Pearl, 2009a,b]:

$$X_i := \theta, \qquad X_j := f_j(\mathbf{PA}_j, U_j) \quad \text{for all } j \neq i. \tag{2.4}$$

The result is a new SCM, whose graph has no arrows into X_i (Figure 2.2) and whose distribution is the *interventional* distribution $P(X_1, \ldots, X_d \mid \mathrm{do}(X_i = \theta))$. Several variables can be set at once. Following Karimi et al. [2020b], this thesis writes $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$ for the intervention that sets every X_i with i in a set of indices $\mathcal{I} \subseteq \{1, \ldots, d\}$ to the matching value θ_i; Pearl [2009b] writes $\mathrm{do}(X = x)$, with X itself a set of variables. Intervening is not the same as conditioning. In the running example, $P(Y \mid X_2 = x_2)$ describes the people who happen to have savings x_2, and those with high savings tend to also have a high income, whereas $P(Y \mid \mathrm{do}(X_2 = x_2))$ describes what happens when savings are set to x_2 and income stays as it was. In algorithmic recourse, the action a person takes is modelled as an intervention on the features they can change [Karimi et al., 2021].



(a) Intervention on X_1 (b) Intervention on X_2

Figure 2.2: The running example (Figure 2.1) under an intervention on income (a) and on savings (b). The intervened variable, drawn as a square, is set from outside: the arrows into it, including the one from its noise variable, are removed, while the arrows out of it stay.

2.1.3 Counterfactual Explanations and Algorithmic Recourse

This subsection introduces counterfactual explanations and algorithmic recourse: what they are, what they are used for and why they matter.

Recourse. Algorithmic recourse has been defined in several overlapping ways. We define it by merging the definitions of Ustun et al. [2019] and Karimi et al. [2022]: a person who receives an unfavourable decision achieves recourse when they can understand a model's decision and act to reverse it by altering actionable input variables. A rejected loan applicant, such as a person in the running example of Section 2.1.2, therefore needs answers to two questions [Karimi et al., 2022]: which profile would have led to the loan being approved (Q1), and which actions would have led to that profile (Q2). An answer to Q1 is a counterfactual explanation and an answer to Q2 is a recommendation of actions. Consider the statement:

> *If I had dropped this glass, it would have broken.*

It describes what would have followed had something been different. A statement of this form is a counterfactual conditional, or counterfactual for short. The antecedent, *I dropped this glass*, is false, and the statement says what would have followed had it been true. Writing D for the antecedent and B for the consequent, *the glass broke*, Lewis [1973] denotes the counterfactual by

$$D \mathrel{\square\!\!\rightarrow} B, \tag{2.5}$$

which we read as "if D were the case, B would be the case". Lewis [1973] interprets this counterfactual in terms of possible worlds, ordered according to their similarity to the actual world. To evaluate Equation 2.5, one compares worlds in which the glass is dropped and breaks with those in which it is dropped but does not break. More precisely, according to Lewis [1973, Analysis 3], the counterfactual is true if some possible world in which the glass is dropped and breaks is more similar to the actual world than every possible world in which it is dropped but does not break. If no possible world makes D true, the counterfactual is vacuously true.

Counterfactual explanations. Wachter et al. [2017] bring Lewis's ideas to machine learning. They define a counterfactual explanation as a statement that gives the feature values behind a decision and the values that would have led to a different decision, with all other features held constant. Many such statements exist for the same decision, and the aim is to find the one that changes the features as little as possible, and so describes the "closest possible world" in which the decision changes, a notion they take from Lewis [1973]. Similarity between worlds thus becomes a distance between feature vectors. Wachter et al. propose the Manhattan (L_1) distance with each feature weighted by the inverse of its median absolute deviation (MAD), which favours explanations that change only a

few features, and they compute the explanation by minimising this distance together with the squared gap between the model's output and the desired output. They also point out that their approach does not rely on knowledge of the causal structure of the world.

Karimi et al. [2022] write the same problem as a constrained optimisation. Let h be a binary classifier, $\mathcal{X}$ the space of feature vectors and $\mathcal{P}(\mathcal{X}) \subseteq \mathcal{X}$ the plausible ones, those consistent with the distribution of training data, and let $\mathbf{x}^{\mathrm{F}}$ be the observed (factual) features of a person. A nearest counterfactual explanation is then

$$\mathbf{x}^{*\mathrm{CFE}} \in \underset{\mathbf{x}^{\mathrm{CFE}} \in \mathcal{P}(\mathcal{X})}{\arg\min}\ \mathrm{dist}(\mathbf{x}^{\mathrm{CFE}}, \mathbf{x}^{\mathrm{F}}) \quad \text{subject to} \quad \begin{array}{l} h(\mathbf{x}^{\mathrm{CFE}}) \neq h(\mathbf{x}^{\mathrm{F}}), \\ \mathbf{x}^{\mathrm{CFE}} = \mathbf{x}^{\mathrm{F}} + \boldsymbol{\delta}, \end{array} \tag{2.6}$$

where $\boldsymbol{\delta}$ shifts each feature independently of the others. Karimi et al. describe these solutions as feature combinations in nearby worlds, again citing Lewis [1973]. A solution of Equation 2.6 answers Q1. It does not answer Q2: although a counterfactual explanation yields a feature vector that would have achieved recourse, the actions that would lead the person to it cannot in general be inferred from it [Karimi et al., 2021].

Structural counterfactuals. An SCM gives a concrete answer to which world is closest to the actual one. Pearl [2009b, Sec. 3.4] reads "had X been x" as a minimal change to the model: the counterfactual world keeps every other structural assignment and the person's own noise values, and changes only what the intervention sets. Pearl [2009a, Sec. 7.4.1] relates this reading to Lewis's counterfactuals. Given the factual features $\mathbf{x}^{\mathrm{F}}$ of a person and an intervention $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$, the counterfactual is computed in three steps [Pearl, 2009a, Thm. 7.1.7], which von Kügelgen et al. [2022] also follow for recourse. In the abduction step, the distribution of the noise is updated with the observation $\mathbf{x}^{\mathrm{F}}$. In the action step, the assignments of the variables in $\mathcal{I}$ are replaced as in Equation 2.4. In the prediction step, the modified assignments are evaluated under the updated noise distribution, which gives a distribution over the counterfactual features [Pearl, 2009a, Sec. 7.1.3]. When only one value of the noise is consistent with $\mathbf{x}^{\mathrm{F}}$, abduction returns that value, $\mathbf{u}^{\mathrm{F}}$, and the counterfactual is a single feature vector. Following von Kügelgen et al. [2022], it is written $\mathbf{x}_{\boldsymbol{\theta}_{\mathcal{I}}}(\mathbf{u}^{\mathrm{F}})$: the features the person would have had, had $\mathbf{X}_{\mathcal{I}}$ been set to $\boldsymbol{\theta}$. In the running example the noise is additive and can be read off the features: $u_1 = x_1$ and $u_2 = x_2 - 0.5\,x_1^2$. A person with $\mathbf{x}^{\mathrm{F}} = (2, 2)$ thus has $u_2 = 0$, and under $\mathrm{do}(X_1 = 3)$ their savings become $0.5 \cdot 3^2 + 0 = 4.5$, so that, with $\mathcal{I} = \{1\}$ and $\boldsymbol{\theta} = 3$, $\mathbf{x}_{\boldsymbol{\theta}_{\mathcal{I}}}(\mathbf{u}^{\mathrm{F}}) = (3, 4.5)$.

Consequential recommendations. Karimi et al. [2021] model an action as an intervention in an SCM of the world in which the person acts, so that its effects on the other variables are taken into account. A recommended set of such actions is called a consequential recommendation [Karimi et al., 2022]. Following von Kügelgen et al. [2022], the action $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$ is written $\boldsymbol{\theta}_{\mathcal{I}}$ for short, and $\mathcal{F}(\mathbf{x}^{\mathrm{F}})$ denotes the set of feasible actions, those that the person can perform. The actions that change the decision are then

$$\mathcal{R} = \left\{ \boldsymbol{\theta}_{\mathcal{I}} \in \mathcal{F}(\mathbf{x}^{\mathrm{F}}) \mid h(\mathbf{x}_{\boldsymbol{\theta}_{\mathcal{I}}}(\mathbf{u}^{\mathrm{F}})) \neq h(\mathbf{x}^{\mathrm{F}}) \right\}, \tag{2.7}$$

and, given the cost $c(\mathbf{x}^{\mathrm{F}}, \boldsymbol{\theta}_{\mathcal{I}})$ of an action for the person, a minimal consequential recommendation is the cheapest of them [Karimi et al., 2022]:

$$\boldsymbol{\theta}^{*}_{\mathcal{I}^{*}} \in \underset{\boldsymbol{\theta}_{\mathcal{I}} \in \mathcal{F}(\mathbf{x}^{\mathrm{F}})}{\arg\min}\, c(\mathbf{x}^{\mathrm{F}}, \boldsymbol{\theta}_{\mathcal{I}}) \quad \text{subject to} \quad h(\mathbf{x}_{\boldsymbol{\theta}_{\mathcal{I}}}(\mathbf{u}^{\mathrm{F}})) \neq h(\mathbf{x}^{\mathrm{F}}). \tag{2.8}$$

Because notation across papers differ, we will now attempt to clarify it. Karimi et al. [2022] write the action as a, the set of feasible actions as $\mathcal{A}(\mathbf{x}^{\mathrm{F}})$, the counterfactual as $\mathbf{x}^{\mathrm{SCF}}(a; \mathbf{x}^{\mathrm{F}})$ and the cost as $\mathrm{cost}(a; \mathbf{x}^{\mathrm{F}})$, whereas von Kügelgen et al. [2022] write the set of feasible actions as $\mathcal{F}(\mathbf{x}^{\mathrm{F}})$. This thesis follows the latter, because a denotes a value of the protected attribute A, and $\mathcal{A}$ could be read as the set of protected attributes. Unlike Equation 2.6, Equation 2.8 requires an SCM. Its solution also gives a counterfactual explanation, $\mathbf{x}_{\boldsymbol{\theta}^{*}_{\mathcal{I}^{*}}}(\mathbf{u}^{\mathrm{F}})$, though not necessarily the nearest one, whereas a solution of Equation 2.6 gives no action. For this reason, Karimi et al. [2022] argue that recourse should primarily be offered through minimal consequential recommendations, and that nearest counterfactual explanations are of secondary importance.

The running example shows the difference. Suppose that the bank approves a loan when the probability of approval is at least 0.5, that is, $h(\mathbf{x}) = \mathbf{1}[x_1 + 0.5\, x_2 \geq 4]$, and take again the person with $\mathbf{x}^{\mathrm{F}} = (2, 2)$, who is rejected because $2 + 0.5 \cdot 2 = 3 < 4$. With the unweighted L_1 distance and no plausibility constraint, the nearest counterfactual explanation raises income by one unit, $\mathbf{x}^{*\mathrm{CFE}} = (3, 2)$. However, raising income to 3 does not lead to this profile. As computed above, savings also rise, to 4.5, so the person ends up at $(3, 4.5)$. The explanation describes a profile that the person cannot reach by changing income alone, and it asks for a larger change than needed. Suppose that the person can change income, savings or both, and that the cost of an action is the total amount by which it changes them. The minimal consequential recommendation then changes income only. If income is set to θ, savings become $0.5\,\theta^2$, and the loan is approved once $\theta + 0.25\,\theta^2 \geq 4$, that is, once θ reaches about 2.47. Raising income by about 0.47 is therefore enough, because savings rise to about 3.06 along with it, whereas the explanation asked for a rise of 1.

2.1.4 Fairness: Statistical and Causal Notions

Machine learning models increasingly inform decisions in high-stakes domains, such as healthcare [Shailaja et al., 2018] and hiring [Raghavan et al., 2020], where their predictions directly affect people's lives. In such domains, predicting well is not enough: the decisions must also be fair to the people they affect. Mitchell et al. [2021] note that a model is now commonly called biased when its predictive performance unjustifiably differs across disadvantaged groups, defined along social axes such as race, gender and class, and describe uncovering and rectifying such biases as the motivation of the field of fairness in machine learning. This subsection presents the two families of fairness criteria that the rest of the thesis relies on: statistical criteria, which compare groups of people, and causal criteria, which ask what the decision would have been had the protected attribute been different.

Statistical criteria. In this subsection the classifier may use the protected attribute A, such as sex or race, so it is applied to all of a person's variables, $\mathbf{v} = (\mathbf{x}, a)$; a classifier that does not use A, as in Section 2.1.3, simply ignores the last entry. Its decision is written $\hat{Y} = h(\mathbf{V})$, with $\mathbf{V} = (\mathbf{X}, A)$. Barocas et al. [2023, Ch. 3] group the statistical criteria into three families, called independence, separation and sufficiency. Each is a conditional independence between A, the outcome Y and the decision; in this order, they require

$$\hat{Y} \perp A, \qquad \hat{Y} \perp A \mid Y, \qquad Y \perp A \mid \hat{Y}. \tag{2.9}$$

For a binary decision, independence requires the approval rate $P(\hat{Y} = 1 \mid A = a)$ to be the same for every value a of A; it is also called demographic parity or statistical parity. Separation requires every group to have the same true positive rate $P(\hat{Y} = 1 \mid Y = 1, A = a)$ and the same false positive rate $P(\hat{Y} = 1 \mid Y = 0, A = a)$. Hardt et al. [2016] call it equalized odds, and call the weaker requirement of equal true positive rates alone equal opportunity. Unlike demographic parity, separation lets the decision depend on A, but only through the outcome Y [Hardt et al., 2016]. Sufficiency requires that people who receive the same decision have the same probability of a positive outcome in every group, that is, equal positive and negative predictive values. The three criteria conflict with one another: for a binary outcome whose rate differs between groups, so that A and Y are dependent, no two of them can hold at the same time except in degenerate cases [Barocas et al., 2023, Ch. 3].

Counterfactual fairness. The criteria above depend only on the joint distribution of A, $\mathbf{X}$, Y and the decision, which is why they are called observational [Kilbertus et al., 2017]. As a result, they cannot say what caused a disparity [Barocas et al., 2023, Ch. 3]. Kilbertus et al. [2017] recall two scenarios, constructed by Hardt et al. [2016], that have intuitively different social interpretations but the same joint distribution, so that no observational criterion can tell them apart. Causal criteria instead reason about an SCM in which A is one of the variables. Kusner et al. [2017] call a decision counterfactually fair if, for every person, it would have been the same in a counterfactual world in which the person belonged to a different demographic group. Formally, given the observed values $\mathbf{x}$ and a, the decision must have the same distribution under $\mathrm{do}(A = a')$ as under $\mathrm{do}(A = a)$, for every value a' of A. The counterfactuals are computed with the three steps of Section 2.1.3, now intervening on A. When only one value of the noise, $\mathbf{u}^{\mathrm{F}}$, is consistent with the observed variables $\mathbf{v}^{\mathrm{F}} = (\mathbf{x}^{\mathrm{F}}, a^{\mathrm{F}})$, the counterfactual is a single vector $\mathbf{v}_a(\mathbf{u}^{\mathrm{F}})$, called the person's counterfactual twin had A been a, and the criterion reads [von Kügelgen et al., 2022]

$$h(\mathbf{v}^{\mathrm{F}}) = h(\mathbf{v}_a(\mathbf{u}^{\mathrm{F}})) \quad \text{for every value } a \text{ of } A \text{ and every person.} \tag{2.10}$$

Leaving A out of the classifier's inputs, which Kusner et al. [2017] call fairness through unawareness, does not ensure this, because other features can carry information about A: in the twin, every feature that A affects changes as well. A classifier that uses only variables that are not descendants of A in the causal graph, on the other hand, is always counterfactually fair [Kusner et al., 2017, Lemma 1].

Path-specific fairness. Counterfactual fairness treats every effect of A on the decision as unfair [Chiappa, 2019], and the most direct way to satisfy it is to discard every descendant of A [Kusner et al., 2017]. Some of these effects may be acceptable, however. In the Berkeley admissions case, discussed by both Kilbertus et al. [2017] and Chiappa [2019], women had a lower admission rate than men overall because they applied more often to competitive departments, and the effect of gender on admission through the choice of department is not regarded as unfair. Kilbertus et al. [2017] call a variable through which A influences the decision in such an accepted way a resolving variable, and say that a variable that is not itself resolving exhibits unresolved discrimination when a directed path from A to it is not blocked by a resolving variable. Nabi and Shpitser [2018] formalise discrimination as the effect of A on the outcome along chosen causal paths, called a path-specific effect, and stress that which paths are discriminatory depends on the domain: a physical fitness test may be appropriate for a construction job but not for an accounting one. They then fit the model to a distribution that is as close as possible to the data distribution while keeping the path-specific effect within bounds that state how much discrimination is tolerated. Chiappa [2019] combines this idea with counterfactual fairness: a decision is path-specific counterfactually fair if it coincides with the one that would have been taken in a counterfactual world in which A had been different along the unfair paths only.

All of these criteria concern the decision itself. Section 2.2.3 reviews how they have been carried over to recourse, where the quantity compared between groups, or between a person and their counterfactual twin, is the cost of the action needed to change the decision [von Kügelgen et al., 2022].

2.1.5 Concept Drift and Distribution Shift

A model is trained on data from the past and applied to people in the present. When the data change in between, a model that was accurate when it was trained can become outdated. Gama et al. [2014] describe this problem, concept drift, as a change over time in the relation between the input data and the target variable. Following them, Webb et al. [2016] call the joint distribution $P_t(\mathbf{X}, Y)$ of the features and the outcome in time window t the concept at time t, and most of the literature now says that concept drift occurs when this distribution changes [Lu et al., 2019]:

$$P_t(\mathbf{X}, Y) \neq P_{t+1}(\mathbf{X}, Y). \tag{2.11}$$

Webb et al. [2016] describe concept drift as the stream-learning counterpart of dataset shift, the name used in batch learning when a model is trained on one dataset and tested on another [Moreno-Torres et al., 2012].

What changes. Writing $P_t(\mathbf{X}, Y) = P_t(\mathbf{X})\, P_t(Y \mid \mathbf{X})$ separates two sources of drift [Lu et al., 2019]. When only $P_t(\mathbf{X})$ changes, the population changes but the relation between features and outcome does not, as when the demographics of a company's customers change over time [Webb et al., 2016]. This does not move the decision boundary, and is also called virtual drift. When $P_t(Y \mid \mathbf{X})$ changes, the relation itself changes, as when a new tax law changes which tax-payers count as

compliant [Webb et al., 2016]. The decision boundary then moves and the model's accuracy drops, which is why this is called actual drift. In practice, the two often occur together [Lu et al., 2019]. For dataset shift, Moreno-Torres et al. [2012] distinguish covariate shift [Shimodaira, 2000], in which $P(\mathbf{X})$ changes while $P(Y \mid \mathbf{X})$ stays the same, prior probability shift, in which $P(Y)$ changes while $P(\mathbf{X} \mid Y)$ stays the same, and concept shift, in which the relation between features and outcome changes. They also tie covariate shift to problems in which the features cause the outcome, and prior probability shift to problems in which the outcome causes the features. Webb et al. [2016] question this restriction, because in many real problems some features cause the outcome, others are caused by it, and others share causes with it. The names also vary: Webb et al. [2016] call a change in $P(Y \mid \mathbf{X})$ class drift, and note that it is also called real concept drift or prior probability shift.

How drift unfolds. Drift also differs in how it unfolds over time. Lu et al. [2019] distinguish four types: in sudden drift, a new concept appears within a short time; in gradual drift, a new concept gradually replaces the old one; in incremental drift, the old concept changes step by step into the new one; and with reoccurring concepts, an old concept returns after some time. They also organise learning under drift into three tasks: detecting whether drift has occurred, understanding it, and adapting the model to it. Understanding drift means finding when it occurs and how long it lasts, how severe it is, and where it occurs, that is, which regions of the data it affects. Methods for detecting and explaining drift are reviewed in Section 2.2.5.

Drift in a causal model. In an SCM, the joint distribution factorises into one mechanism per variable (Equation 2.2), and by the principle of independent mechanisms one of them can change while the others stay the same (Section 2.1.2). Schölkopf et al. [2021] go further with the sparse mechanism shift hypothesis: small changes in the distribution tend to affect only a few mechanisms, rather than all of them at once. Drift can then be described by which mechanisms changed. This turns the question of where drift occurs, which Lu et al. [2019] answer with regions of the data, into a question about the causal model, and the categories above do not answer it: a change in $P(\mathbf{X})$ can come from any of the mechanisms of the features.

The running example shows why this matters for recourse. Suppose that people start saving less, so that the equation of savings becomes $X_2 := 0.4\,X_1^2 + U_2$, while the equations of income and of the decision stay the same. Since the equation of Y is unchanged, so is $P(Y \mid X_1, X_2)$: for the classifier this is covariate shift, and h needs no update. The consequential recommendation of Section 2.1.3, however, no longer works. For the person with $\mathbf{x}^F = (2, 2)$, abduction now gives $u_2 = 2 - 0.4 \cdot 4 = 0.4$. Raising income to 2.47 now raises savings only to about 2.84, so the score $2.47 + 0.5 \cdot 2.84 \approx 3.89$ stays below 4 and the loan is still rejected; under the new equation, income has to reach about 2.52. Recourse computed with an outdated causal model can thus fail even when the classifier itself is still valid. Recourse under drift, and methods that learn causal structure from changes in

the distribution, are reviewed in Section 2.2.4.

Drift can also undo fairness. Deho et al. [2024] find that covariate drift is common in real data and can substantially reduce the fairness of models trained to be fair, and that the size and direction of the drift do not reliably predict how much fairness is lost. Fairness under distribution shift is reviewed in Section 2.2.6.

2.1.6 Splines and Additive Models

2.1.7 Kolmogorov–Arnold Networks

2.2 Literature Review

This chapter reviews the work on which a causal, fair and drift-aware recourse method can build. Section 2.2.1 covers how counterfactuals are searched for, and Section 2.2.2 how they are kept plausible and causally consistent. Section 2.2.3 asks whether recourse is fair, and Section 2.2.4 examines what happens when models, populations or causal mechanisms change over time. Finally, Section 2.2.8 turns to readable representations of structural equations, including those based on Kolmogorov–Arnold Networks (KANs).

2.2.1 Counterfactual Search Methods

In this section, we analyse the current landscape of counterfactual search methods, focusing on which strategies are used to find valid counterfactuals and which trade-offs they make between search efficiency, plausibility and causal consistency. We group the methods into three categories: exact solver-based methods, gradient-based methods, and evolutionary and surrogate-based methods.

2.2.1.1 Exact Solver-Based Methods

Russell [2019] proposes a mixed-integer programming approach to generating counterfactual explanations for linear classifiers, including logistic regression, linear regression and support vector machines. The search is formulated as an optimisation problem that minimises a weighted ℓ_1 distance between the original instance and the counterfactual, which keeps changes sparse (i.e., as few features as possible are changed). Its main contribution is a set of constraints, called a mixed polytope, that guarantees coherent counterfactuals: solutions that map back onto the underlying data structure, even when a variable can take either a continuous value or one of several discrete states. Diversity is obtained by restricting the variables altered in previously generated counterfactuals. Results on the LSAT and FICO datasets show that the method finds counterfactuals that are close to the original instance and cross the model's decision boundary, while avoiding the brute-force enumeration required by earlier approaches. However,

the approach is designed for linear models, so it does not extend directly to complex neural networks. It is also blind to causal relationships between features: it guarantees coherence, not causal consistency. Finally, explanations are presented as short text statements, and the author notes that a complete set can be overwhelming and that more work is needed on how to present and navigate it interactively.

Kanamori et al. [2020] also rely on mathematical optimisation, using mixed-integer linear optimisation for linear models and tree ensembles, but do so in a distribution-aware manner. Their cost function combines the Mahalanobis distance, which accounts for feature correlations, with the local outlier factor, which avoids counterfactuals in low-density regions. However, this comes at a computational price: on the FICO dataset with a random forest, DACE took 519 seconds on average under a 600-second limit, against 22 to 34 seconds for the baselines. Moreover, although it encourages counterfactuals not to break correlations, it does not guarantee causal consistency.

Karimi et al. [2020a] propose MACE, a model-agnostic method that generates counterfactuals using Satisfiability Modulo Theories (SMT) solvers. The model, the distance function and the constraints are all translated into logic formulae, and additional constraints make the counterfactuals plausible and diverse. The method finds a counterfactual for every individual considered (100% coverage) at provably optimal distances. The problem is that the model must be expressible in first-order logic, which may not be feasible for all types of models, especially complex ones. Once again, the method is blind to causal relationships between features, and it can be computationally expensive because its running time depends directly on the solver: under five seconds for logistic regression, but between one minute and three hours for a multilayer perceptron.

2.2.1.2 Gradient-Based Methods

Van Looveren and Klaise [2020] use class prototypes to guide the counterfactual search. A term added to the loss function encourages the counterfactual to be close to the prototype of the desired class, obtained either from an encoder or from class-specific k-d trees, which makes the generated counterfactuals more interpretable and keeps them closer to the data manifold. Evaluation on the MNIST and Breast Cancer Wisconsin datasets shows that the method generates counterfactuals that are close to the original instance and cross the decision boundary, and that it is faster than searches without prototypes. It also removes the bottleneck of numerical gradient evaluation for black-box models. Notwithstanding, the method relies heavily on the quality of the prototypes and is, again, blind to causal relationships between features. Matching a prototype can also lead to less sparse counterfactuals, as many features may have to change, and the authors report a trade-off between sparsity and interpretability.

Mothilal et al. [2020] propose DiCE, a framework that generates multiple, diverse and actionable counterfactuals for machine learning classifiers. It moves explainability from showing a single path of recourse to offering individuals a variety of options to choose from. DiCE initialises k candidates randomly and optimises

them using a determinantal point process (DPP) term to ensure diversity, along with proximity to the original instance. The novelty of the search mechanism is that it optimises all candidates jointly. The framework is evaluated on the COMPAS, Adult-Income, German-Credit and Lending Club datasets. Its limitations are that it assumes a differentiable and static model, and that it varies features independently, which can produce causally infeasible counterfactuals; causal constraints are only used to filter these out afterwards.

2.2.1.3 Evolutionary and Surrogate-Based Methods

White and d'Avila Garcez [2019] introduce CLEAR and a definition of the fidelity of local explanation models to the underlying classifier, based on distances to a decision boundary. The method generates b-counterfactuals, which state the minimum changes necessary to flip the classifier's prediction, and uses them to measure and improve the fidelity of local regression models. Across five case studies, CLEAR improves on the fidelity of LIME by an average of over 40%. However, its neighbourhood datasets do not always represent the relevant regions of the feature space, fidelity varies considerably between datasets, and b-counterfactuals that change several features are left as future work.

Dandl et al. [2020] propose MOC which, to the best of the authors' knowledge, is the first method to formalise the counterfactual search as a multi-objective optimisation problem. It adapts the NSGA-II algorithm to return a set of Pareto-optimal counterfactuals with different trade-offs between four objectives: reaching the desired prediction, staying close to the original instance, changing few features and staying close to the observed data. The method is model-agnostic and can be applied to any black-box model. Nevertheless, the benchmark is limited to binary classification on datasets with at most 40 features, which raises concerns about scalability to high-dimensional data. The work also leaves open how one should choose the final counterfactual from the Pareto front, gives no guarantee of causal consistency, and can be computationally expensive due to the nature of evolutionary algorithms.

2.2.2 Plausibility and Causal Consistency

A counterfactual that is close to the original instance is not necessarily one that a person can reach. This section reviews two responses to that problem: keeping counterfactuals on the data manifold, and explicitly modelling the causal relationships between features.

2.2.2.1 Data-Manifold Methods

Poyiadzi et al. [2020] propose FACE, a graph-based method for generating counterfactuals. It constructs a graph where nodes are observed data points (e.g., people's financial profiles), edges connect nearby nodes, and density-weighted edge weights represent the cost of moving from one node to another, which favours

paths through high-density regions. The method then searches for the shortest path from the original instance to a node of the desired class that satisfies prediction and density thresholds. On a synthetic dataset and on MNIST, the authors show that the method generates not only counterfactuals but also intermediate steps that explain how to reach them. However, the method does not guarantee that real-world actions will succeed: it ensures plausibility, which should not be confused with causality, nor with proof that an individual can take that set of actions. Evaluation on real-world data of a dynamic nature is left as future work.

Olson et al. [2021] introduce counterfactual state explanations for deep reinforcement learning agents via generative deep learning. An encoder maps the game state to a latent representation, which is fed together with the agent's action distribution into a generator. The result is the minimal visual modification that makes the agent choose a different action. The networks are trained using a combination of adversarial and reconstruction losses, so that the counterfactual states are both realistic and close to the original state. Two user studies on Atari games with non-experts show that (i) the generated states have sufficient fidelity to the actual game, and (ii) they help participants identify a flawed agent, which the authors deliberately blinded to its own ship's position, more effectively than a nearest-neighbour baseline or no explanation at all. The method could therefore serve as an auditing tool to identify agent misbehaviour before real-world deployment. Beyond being restricted to reinforcement learning agents and evaluated on Atari games rather than on high-stakes real-world problems, the method does not explain the long-term effects of the chosen action on the environment, so no temporal consistency is guaranteed. The generative model also has no concern for causal consistency, which increases the risk of providing misleading explanations.

Joshi et al. [2019] introduce REVISE, a method for plausible algorithmic recourse through generative models. A variational autoencoder (VAE) encodes a person into a latent space, where gradient descent pushes the candidate towards the desired label while keeping it close to the original instance. Recourse is the difference between the original and the generated instance. Identified limitations include that the suggested changes are not sparse, which may make them difficult for a person to act upon; that the reported evaluation is mainly qualitative, with quantitative results relegated to the appendix; and that realism depends on the VAE. Although the authors use a method (CEVAE) that captures the effect of a treatment on the outcome adjusted for confounders (variables that influence both the treatment and the outcome), they do not model causal relationships between features, so recourse is still a list of independent feature changes, and they provide no guarantee that it works. They also show that recourse can act as a diagnostic, revealing whether a classifier is systematically confounded by specific attributes.

2.2.2.2 Causal Constraints and Recourse as Interventions

Mahajan et al. [2019] are among the first to connect the feasibility of counterfactuals with causal relationships between features, arguing that feasibility is fun-

damentally a causal concept. They use a VAE-based generator with constraints derived from a partial structural causal model (SCM) and, when such constraints cannot be expressed, learn feasibility from users' binary feedback. However, causal relationships are soft constraints and are therefore not guaranteed to be satisfied, the output is still a counterfactual state rather than a set of actions, and the comparison covers only one external baseline.

Karimi et al. [2021], in contrast, present a method that outputs actions as interventions on an SCM, thus guaranteeing causal consistency when the SCM is correct. They prove that acting on a nearest counterfactual explanation is only guaranteed to achieve recourse when the acted-upon variables have no descendants, and that a minimal intervention never costs more than a counterfactual-based action that achieves recourse. The method extends the aforementioned MACE framework and distinguishes immutable, mutable but non-actionable, and actionable features, successfully reframing recourse from a nearest-point search to actions and their consequences. Regarding limitations, the method requires the true SCM, with no hidden confounders and invertible equations, and the experiments are limited to small graphs, with little robustness analysis. The authors leave open (i) actions that affect multiple features, (ii) sequential actions and (iii) the choice of cost functions.

Karimi et al. [2020b] tackle the weakness of having to know the true SCM. They start by showing that two causal models that are observationally equivalent can lead to different recourse recommendations, so unless we know the true structural equations, we cannot guarantee that recourse is valid, no matter how much data is available. They then propose two probabilistic methods for the case where the structural equations are unknown but the causal graph is known: individual recourse using a Gaussian process SCM (GP-SCM), and subpopulation recourse based on the conditional average treatment effect (CATE). The first assumes that each structural equation is a function of its parents plus Gaussian noise, yielding a distribution over a person's counterfactuals. The second drops all assumptions about the structural equations and estimates the average effect of the action on people who share the same non-descendant features. The limitations are that (i) the causal graph is still required, as only the equations are relaxed, (ii) the additive Gaussian noise assumption may not hold under the true SCM, leading to sub-optimal solutions, (iii) the experiments are small, and (iv) subpopulation recourse only ensures that the average effect of the action is positive for a group of people, not that it works for a given individual. The authors treat the classifier as given, and they point to fair decision making with a classifier that evolves over time as an important direction for future research.

2.2.3 Fairness of Recourse

Even when recourse is valid, it may be harder for some people to achieve than for others. This section reviews how that inequality has been measured, first through distance-based audits, then through causal notions of fairness, and finally over time.

Sharma et al. [2020] propose CERTIFAI, a framework that uses counterfactual explanations generated by a genetic algorithm as a model-agnostic way of auditing machine learning models for fairness, robustness and interpretability. They introduce CERScore, a black-box robustness metric defined as the average distance between inputs and their counterfactuals, and burden, the expected distance to the counterfactual for a group, which allows groups to be compared. The main limitations are that a genetic approach does not guarantee optimality, that actionability depends on feature constraints set by the user, and that causal consistency is not considered.

von Kügelgen et al. [2022] make recourse a fairness criterion and connect it formally to counterfactual fairness. They define group-level and individual-level notions of fair causal recourse and show that: (i) group-level fairness does not imply individual-level fairness; (ii) a counterfactually fair classifier does not guarantee fair recourse (fair prediction and fair recourse are complementary, not substitutes); and (iii) if the classifier only uses features that are non-descendants of the protected attribute, and the feasible actions and their costs do not change under a counterfactual change of that attribute, then recourse is individually fair. They also show that non-causal metrics misrepresent recourse unfairness, and they detect recourse discrimination in a case study on the Adult dataset. Regarding limitations, the individual-level notion needs a counterfactual on the protected attribute, which requires the full SCM; the fix based on non-descendant features can lead to a drop in accuracy; constraining training directly is difficult because the search over intervention targets is combinatorial; and the individual-level measure is a maximum over individuals, so it may be dominated by outliers. As changing the classifier does not address the root of the problem, the authors also propose societal interventions on the data-generating process as an alternative.

Fonseca et al. [2023] move the question to a dynamic setting. In agent-based simulations, they define a recourse reliability metric and study two effects: competition with other agents acting on recourse, and competition with new agents entering the environment. Only a small set of parameter settings yields recourse that is reliable over time. However, these results rest on simulated adaptation functions, which only approximate human behaviour, and no causal model is used.

Bell et al. [2025] build on this simulation work and define two new metrics, grounded in substantive equality of opportunity, for measuring fairness in this context: Effort-to-Recourse (ETR) and Time-to-Recourse (TTR). Disadvantaged groups need more of both, even under a fair classifier. The proposed Circumstance-Normalized Selection (CNS) adjusts selection after scoring to reduce that gap. In terms of limitations, the work relies on agent simulations, so its results depend on assumed human behaviour, it uses no causal models, and the experiments use a simple logistic classifier. The authors also point to the lack of public real-world recourse data as a main limitation.

2.2.4 Recourse and Causal Models Under Drift

Most of the works above assume that the model, the population and the causal mechanisms stay fixed. This section relaxes that assumption in three steps: the world changes and invalidates recourse, recourse itself causes change, and causal structure has to be learned from change.

2.2.4.1 Recourse in a Changing World

One of the most relevant works on recourse in changing environments is De Toni et al. [2025], where the authors treat the world as a causal process that changes over time, rather than treating the model as the only shifting component. They start by proving that counterfactual recourse for a future time cannot be optimal unless the noise has zero variance, and that robust uncertainty sets are not enough (i.e., searching for a counterfactual and giving it an error margin does not counteract time). Their solution, T-SAR, gives subpopulation recourse using an estimator of the stochastic process. Notwithstanding, the estimator is fitted once and the classifier is also trained once, so trends in the classifier and in the cost are not studied, and the results depend on the quality of the estimator. The work covers one class of time-series SCM (TiMINo), does not identify which mechanism changed, and has no fairness analysis.

Kostrzewa et al. [2026] repair counterfactuals as the classifier updates, instead of making a prediction about the future, keeping a set of counterfactuals valid and plausible across time. A counterfactual is repaired when it becomes invalid or its target-class confidence is low, and it is retired once the updated classifier already gives the original instance the target class. Repairing is cheaper than regenerating, although repaired counterfactuals can move further from the original instance. The main limitation of this work is that it is a proof of concept: it uses a local linear surrogate, does not model causal relationships, has no fairness mechanisms, and covers only gradual drift on low-dimensional synthetic datasets.

Upadhyay et al. [2021] propose ROAR, a framework that hardens recourse in advance against a bounded model shift through a minimax objective, and prove that the extra cost of this robustness is bounded by a 1-Lipschitz function. Unfortunately, the framework applies only to linear models (non-linear models are first approximated using LIME) and to non-causal recourse. It also assumes bounded shifts rather than continuous drift.

Rawal et al. [2020] give this problem a theoretical basis. They establish a lower bound on the probability that recourse is invalidated by a model update, and show a trade-off between that probability and recourse cost. Experiments on temporal, geospatial and data-correction shifts confirm that invalidation happens in practice, and that causally generated recourse is not immune to it. The work offers no solution, however, and its theory covers only sparse counterfactual recourse.

2.2.4.2 Recourse That Causes Its Own Drift

König et al. [2025] define performative validity as whether recourse still holds under the best model after people act on it. Their work proves that recourse can fail when actions intervene on, or are influenced by, effect variables. Improvement-focused causal recourse stays valid across a broad range of settings, whereas standard counterfactual and causal recourse often do not. The main limitations are that the work considers a single step, without repeated applications, that it assumes a single, static SCM, and that it requires causal knowledge.

Liu et al. [2025], on the other hand, follow the loop over many rounds with a limited number of positive decisions. Recourse pushes decision standards up, raising cost and failure rates. Fair-top-k and Dynamic Continual Learning reduce the damage, with recourse costs falling by around 70%: the former encourages diversity in the feature space by favouring people from less crowded regions, and the latter reduces how sharply the model changes. Here, there is no causal model, and the theory focuses on logistic regression in a saturated system. In both papers, drift comes from recourse and not from the outside world.

2.2.4.3 Learning Causal Structure from Change

Huang et al. [2020] propose Constraint-based causal Discovery from heterogeneous/NOnstationary Data (CD-NOD), a framework that leverages distribution shifts to learn causal structure. It also extracts a low-dimensional "driving force" of the change and allows edges to appear and vanish over time. Perry et al. [2022] make sparsity explicit through the sparse mechanism shift hypothesis and propose the Mechanism Shift Score, proving that the full graph can be identified with high probability if the hypothesis holds. Although promising, both works are limited in scale: they are batch methods, so neither fits a data stream directly, and the kernel-based tests used by CD-NOD have a cost of $O(N^3)$ in the number of samples. Both also rule out most hidden confounders, sparse shift remains a hypothesis rather than an established fact, and neither work produces counterfactuals or recourse.

2.2.4.4 Tracking Causal Models Online

2.2.5 Detecting and Explaining Concept Drift

2.2.6 Fairness Under Distribution Shift

2.2.7 Interpretable Additive Models with Interactions

2.2.8 Readable Structural Equations

Ma et al. [2025] propose Causal-KAN, which puts SCM functions into KANs and encodes causal direction through initialisation and constraints, claiming less overfitting and more readable mappings. Alongside it, Almodóvar et al. [2026] show that KANs can be used to learn the structural equations of a known causal graph, so that the learned mechanisms can be inspected directly. Their method assumes causal sufficiency, and the authors note that it cannot guarantee that the learned mechanisms coincide with the true structural equations.

Additive structural equations also have a statistical precursor in Bühlmann et al. [2014], whose causal additive models (CAM) separate the search for a variable order from the sparse selection of edges, with consistency results in both low and high dimensions. CAM assumes that there are no hidden variables or cycles, and the authors note that such models are not closed under marginalisation, so a hidden variable can take the data outside the model class.

Nevertheless, none of these works provides any connection to recourse, much less to drift and fairness.

2.2.9 Comparison and Research Gaps

Work	Rec.	Causal	Fair	Time	Loc.	Read.	Unc.	Online
Counterfactual search								
Russell [2019]	✓							
Kanamori et al. [2020]	✓							
Karimi et al. [2020a]	✓							
Van Looveren and Klaise [2020]	✓							
Mothilal et al. [2020]	✓	(✓)						
White and d'Avila Garcez [2019]	✓							
Dandl et al. [2020]	✓							
Plausibility and causal consistency								
Poyiadzi et al. [2020]	✓							
Olson et al. [2021]	✓							
Joshi et al. [2019]	✓	(✓)						
Mahajan et al. [2019]	✓	(✓)						
Karimi et al. [2021]	✓	✓						
Karimi et al. [2020b]	✓	✓					✓	
Fairness of recourse								
Sharma et al. [2020]	✓		✓					
von Kügelgen et al. [2022]	✓	✓	✓					
Fonseca et al. [2023]	✓			✓				
Bell et al. [2025]	✓		✓	✓				
Recourse and causal models under drift								
De Toni et al. [2025]	✓	✓		✓			(✓)	
Kostrzewa et al. [2026]	✓			✓				(✓)
Upadhyay et al. [2021]	✓			(✓)				
Rawal et al. [2020]	✓	(✓)		✓				
König et al. [2025]	✓	✓		(✓)				
Liu et al. [2025]	✓		(✓)	✓				
Huang et al. [2020]		✓		✓	✓			
Perry et al. [2022]		✓		(✓)	✓			
Readable structural equations								
Ma et al. [2025]		✓				✓		
Almodóvar et al. [2026]		✓				✓		
Bühlmann et al. [2014]		✓				✓		
This work	○	✓	○	✓	✓	✓	✓	○

Table 2.2: Comparison of the reviewed works. **Rec.**: produces counterfactuals or recourse actions; **Causal**: uses a causal graph or structural causal model; **Fair**: measures or enforces fairness; **Time**: handles change over time (model, population or mechanism); **Loc.**: identifies which mechanism changed; **Read.**: learned mechanisms can be read directly; **Unc.**: quantifies uncertainty; **Online**: updates from a data stream. ✓ addressed; (✓) partly addressed; ○ planned in this work; blank: not addressed.

Chapter 3

Proposed Approach

3.1 Problem Formulation

3.2 Candidate Architectures

3.3 Experimental Design

3.4 Summary

Chapter 4

Conclusion

References

Alejandro Almodóvar, Mar Elizo, Patricia A. Apellániz, Santiago Zazo, and Juan Parras. Kolmogorov-Arnold causal generative models. arXiv:2603.20184, 2026.

Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, Cambridge, MA, 2023.

Andrew Bell, Joao Fonseca, Carlo Abrate, Francesco Bonchi, and Julia Stoyanovich. How much effort is enough? fairness in algorithmic recourse through the lens of substantive equality of opportunity. In *Proceedings of the 5th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 170–184, 2025.

Peter Bühlmann, Jonas Peters, and Jan Ernest. Cam: Causal additive models, high-dimensional order search and penalized regression. *The Annals of Statistics*, 42(6):2526–2556, 2014.

Silvia Chiappa. Path-specific counterfactual fairness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 7801–7808, 2019. doi: 10.1609/aaai.v33i01.33017801.

Susanne Dandl, Christoph Molnar, Martin Binder, and Bernd Bischl. Multi-objective counterfactual explanations. In *Parallel Problem Solving from Nature – PPSN XVI*, pages 448–469. Springer, 2020. doi: 10.1007/978-3-030-58112-1_31.

Giovanni De Toni, Stefano Teso, Bruno Lepri, and Andrea Passerini. Time can invalidate algorithmic recourse. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, pages 89–107, 2025.

Oscar Blessed Deho, Michael Bewong, Selasi Kwashie, Jiuyong Li, Jixue Liu, Lin Liu, and Srecko Joksimovic. Is it still fair? A comparative evaluation of fairness algorithms through the lens of covariate drift. arXiv:2409.12428, 2024.

João Fonseca, Andrew Bell, Carlo Abrate, Francesco Bonchi, and Julia Stoyanovich. Setting the right expectations: Algorithmic recourse over time. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 1–11, 2023.

João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46 (4):44:1–44:37, 2014. doi: 10.1145/2523813.

Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, volume 29, pages 3315–3323, 2016.

Biwei Huang, Kun Zhang, Jiji Zhang, Joseph Ramsey, Ruben Sanchez-Romero, Clark Glymour, and Bernhard Schölkopf. Causal discovery from heterogeneous/nonstationary data. *Journal of Machine Learning Research*, 21(89):1–53, 2020.

Shalmali Joshi, Oluwasanmi Koyejo, Warut Vijitbenjaronk, Been Kim, and Joydeep Ghosh. Towards realistic individual recourse and actionable explanations in black-box decision making systems. arXiv:1907.09615, 2019.

Kentaro Kanamori, Takuya Takagi, Ken Kobayashi, and Hiroki Arimura. DACE: Distribution-aware counterfactual explanation by mixed-integer linear optimization. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 2855–2862, 2020. doi: 10.24963/ijcai.2020/395.

Amir-Hossein Karimi, Gilles Barthe, Borja Balle, and Isabel Valera. Model-agnostic counterfactual explanations for consequential decisions. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 108 of *Proceedings of Machine Learning Research*, pages 895–905, 2020a.

Amir-Hossein Karimi, Julius von Kügelgen, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse under imperfect causal knowledge: A probabilistic approach. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020b.

Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counterfactual explanations to interventions. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 353–362, 2021.

Amir-Hossein Karimi, Gilles Barthe, Bernhard Schölkopf, and Isabel Valera. A survey of algorithmic recourse: Contrastive explanations and consequential recommendations. *ACM Comput. Surv.*, 55(5), December 2022. ISSN 0360-0300. doi: 10.1145/3527848. URL `https://doi.org/10.1145/3527848`.

Niki Kilbertus, Mateo Rojas-Carulla, Giambattista Parascandolo, Moritz Hardt, Dominik Janzing, and Bernhard Schölkopf. Avoiding discrimination through causal reasoning. In *Advances in Neural Information Processing Systems*, volume 30, pages 656–666, 2017.

Gunnar König, Hidde Fokkema, Timo Freiesleben, Celestine Mendler-Dünner, and Ulrike von Luxburg. Performative validity of recourse explanations. *Advances in Neural Information Processing Systems*, 38:139334–139370, 2025.

Marcin Kostrzewa, Jerzy Stefanowski, and Maciej Zięba. Counterfactual explanations under concept drift. *arXiv preprint arXiv:2605.17651*, 2026.

Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, volume 30, pages 4066–4076, 2017.

David Lewis. Counterfactuals and comparative possibility. In *IFS: Conditionals, belief, decision, chance and time*, pages 57–85. Springer, 1973.

Bo-Yi Liu, Zhi-Xuan Liu, Kuan Lun Chen, Shih-Yu Tsai, Jie Gao, and Hao-Tsung Yang. Understanding endogenous data drift in adaptive models with recourse-seeking users. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 8, pages 1598–1610, 2025.

Christos Louizos, Uri Shalit, Joris M. Mooij, David Sontag, Richard Zemel, and Max Welling. Causal effect inference with deep latent-variable models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Jie Lu, Anjin Liu, Fan Dong, Feng Gu, João Gama, and Guangquan Zhang. Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, 31(12):2346–2363, 2019. doi: 10.1109/TKDE.2018.2876857.

Yiming Ma, Yanmin Li, Kaiyue Liu, Zhihao Chen, Lihua Liu, Hongbin Huang, and Hang Zhang. Causal-kan: Learning causal mapping representations by injecting causal knowledge. In *2025 11th International Conference on Big Data and Information Analytics (BigDIA)*, pages 613–620. IEEE, 2025.

Divyat Mahajan, Chenhao Tan, and Amit Sharma. Preserving causal constraints in counterfactual explanations for machine learning classifiers. arXiv:1912.03277, 2019.

Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, and Kristian Lum. Algorithmic fairness: Choices, assumptions, and definitions. *Annual Review of Statistics and Its Application*, 8:141–163, 2021. doi: 10.1146/annurev-statistics-042720-125902.

Jose G. Moreno-Torres, Troy Raeder, Rocío Alaiz-Rodríguez, Nitesh V. Chawla, and Francisco Herrera. A unifying view on dataset shift in classification. *Pattern Recognition*, 45(1):521–530, 2012. doi: 10.1016/j.patcog.2011.06.019.

Ramaravind K. Mothilal, Amit Sharma, and Chenhao Tan. Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 607–617, 2020.

Razieh Nabi and Ilya Shpitser. Fair inference on outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 1931–1940, 2018.

Matthew L. Olson, Roli Khanna, Lawrence Neal, Fuxin Li, and Weng-Keen Wong. Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence*, 295:103455, 2021. ISSN 0004-3702. doi: 10.1016/j.artint.2021.103455. URL `http://dx.doi.org/10.1016/j.artint.2021.103455`.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, second edition, 2009a. doi: 10.1017/CBO9780511803161.

Judea Pearl. Causal inference in statistics: An overview. *Statistics Surveys*, 3: 96–146, 2009b. doi: 10.1214/09-SS057.

Ronan Perry, Julius Von Kügelgen, and Bernhard Schölkopf. Causal discovery in heterogeneous environments under the sparse mechanism shift hypothesis. *Advances in Neural Information Processing Systems*, 35:10904–10917, 2022.

Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge, MA, 2017.

Rafael Poyiadzi, Kacper Sokol, Raul Santos-Rodriguez, Tijl De Bie, and Peter Flach. FACE: Feasible and actionable counterfactual explanations. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 344–350, 2020.

Manish Raghavan, Solon Barocas, Jon Kleinberg, and Karen Levy. Mitigating bias in algorithmic hiring: Evaluating claims and practices. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*, pages 469–481, 2020.

Kaivalya Rawal, Ece Kamar, and Himabindu Lakkaraju. Algorithmic recourse in the wild: Understanding the impact of data and model shifts. arXiv:2012.11788, 2020.

Chris Russell. Efficient search for diverse coherent explanations. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 20–28, 2019. doi: 10.1145/3287560.3287569.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. doi: 10.1109/JPROC.2021.3058954.

K Shailaja, Banoth Seetharamulu, and MA Jabbar. Machine learning in healthcare: A review. In *2018 Second international conference on electronics, communication and aerospace technology (ICECA)*, pages 910–914. IEEE, 2018.

Shubham Sharma, Jette Henderson, and Joydeep Ghosh. CERTIFAI: A common framework to provide explanations and analyse the fairness and robustness of black-box models. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 166–172, 2020.

Hidetoshi Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244, 2000. ISSN 0378-3758. doi: https://doi.org/10.1016/S0378-3758(00)00115-4. URL `https://www.sciencedirect.com/science/article/pii/S0378375800001154`.

Sohini Upadhyay, Shalmali Joshi, and Himabindu Lakkaraju. Towards robust and reliable algorithmic recourse. *Advances in Neural Information Processing Systems*, 34:16926–16937, 2021.

Berk Ustun, Alexander Spangher, and Yang Liu. Actionable recourse in linear classification. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 10–19, 2019.

Arnaud Van Looveren and Janis Klaise. Interpretable counterfactual explanations guided by prototypes. arXiv:1907.02584, 2020.

Julius von Kügelgen, Amir-Hossein Karimi, Umang Bhatt, Isabel Valera, Adrian Weller, and Bernhard Schölkopf. On the fairness of causal algorithmic recourse. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 9584–9594, 2022. doi: 10.1609/aaai.v36i9.21192.

Sandra Wachter, Brent Mittelstadt, and Chris Russell. Counterfactual explanations without opening the black box: Automated decisions and the gdpr. *Harv. JL & Tech.*, 31:841, 2017.

Geoffrey I. Webb, Roy Hyde, Hong Cao, Hai Long Nguyen, and François Petitjean. Characterizing concept drift. *Data Mining and Knowledge Discovery*, 30(4): 964–994, 2016. doi: 10.1007/s10618-015-0448-4.

Adam White and Artur d'Avila Garcez. Measurable counterfactual local explanations for any classifier. arXiv:1908.03020, 2019.

Appendices

Appendix A

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Appendix B

Title for the second appendix

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